

TB Multiband Compressor

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Overview

Controls only the frequency areas that need help, so lows, body, presence, and air can move independently without forcing the whole signal through permanent crossover bands.

What this plug-in solves

A full-band compressor can solve a level problem while also pulling down frequencies that were already balanced. Traditional crossover multiband processing can make every part of the spectrum pass through a split even when only one area needs help. TB Multiband Compressor lets musicians, producers, and sound designers place up to four independent bands only where a problem or opportunity exists, leaving the rest of the signal on its direct path.

What you can expect

- Independent control of low end, body, presence, and air without forcing every frequency through a permanent crossover split
- Compression or expansion that can reduce a problem, lift detail, create motion, or react to an external key
- Stereo, Mid, or Side targeting for focused center control and deliberate width shaping
- A direct graph, band-only DELTA monitoring, factory starting points, and repeatable User preset recall

How it works

Each active band listens to a chosen frequency area and changes only that area's contribution. When a band is inactive or its detector calls for no movement, the rest of the signal remains on the direct path rather than being forced through a permanent full-spectrum split.

COMP or EXPAND selects which side of Threshold becomes active. Range then chooses whether the active band moves down or up and limits the size of that move. This separation makes gentle control, upward detail, rhythmic ducking, and below-threshold shaping possible with the same band controls.

Quick start

- 1** Insert the plug-in on a track or bus, load Gentle Balance or Broad Control, and leave only the bands you need enabled.
- 2** Select a band and place Frequency over the problem or detail you want to shape; set WIDTH so the colored area covers only the useful region.
- 3** Choose COMP for events above Threshold or EXPAND for events below it, then set negative Range for reduction or positive Range for lift.
- 4** Move Threshold until the red activity contour follows the intended events rather than staying active all the time.
- 5** Shape the feel with Ratio, Knee, Attack, and Release while listening in the complete arrangement.
- 6** Use DELTA briefly to confirm the band is affecting the intended material, then turn DELTA off.
- 7** Choose Stereo, Mid, or Side and set Stereo Link when the band should preserve coordinated left/right movement.
- 8** Set DYNAMIC or LINEAR, Oversampling, Lookahead, and Mix last; compare with BYPASS at a similar loudness and with DAW delay compensation enabled.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

ANALYZER

The large display shows the main input spectrum. It does not show the output or external sidechain. Numbered nodes identify the four band slots; colored regions and full-height edges show each band's center and width. The pale contour shows the maximum change allowed by Range, while the thinner red contour shows the gain movement currently being applied. These contours are control and activity guides, not a conventional output-EQ measurement.

Direct band editing

Click a node to select its band. Drag left or right to change Frequency and up or down to change Range. Drag a colored edge, or use the wheel over a node, to change WIDTH. Double-click empty graph space to enable the first unused band at that frequency. These gestures edit the same controls shown in the selected-band panel below the graph.

Compression, expansion, and signed Range

COMP reacts when the selected frequency becomes louder than Threshold; EXPAND reacts when it becomes quieter. Range separately chooses the direction and the maximum movement. A negative Range reduces the band and a positive Range lifts it, so COMP and EXPAND can each create either control or emphasis. Ratio, Knee, Attack, and Release shape how that movement begins and returns.

Sidechain and stereo target

SIDECHAIN Off listens to the main signal; On listens to the DAW's shared auxiliary key input for the selected band. Every externally keyed band hears the same physical key bus but still reacts in its own frequency area. Stereo processes the complete stereo image, Mid targets centered content, and Side targets the width information. Stereo Link coordinates left and right movement in Stereo mode and is intentionally inactive for Mid or Side.

DELTA monitoring

DELTA plays only the selected band's wet-minus-dry contribution, making it easy to hear what that band removes or adds. Click DELTA to latch it; clicking the same band again turns it off, while selecting DELTA on another band transfers the monitor. Right-button hold on an enabled node gives a temporary DELTA listen, and horizontal movement during the hold changes that band's Frequency. DELTA is a temporary panel monitor: it is not automated, saved, or recalled, and final BYPASS always takes priority.

Processing quality and timing

DYNAMIC is the immediate, zero-latency-capable path when Oversampling is Off and Lookahead is at its neutral position. LINEAR uses a linear-phase band path and adds more processing delay. Oversampling can clean up fast gain movement at the cost of CPU and additional delay. Lookahead helps the detector catch sudden peaks but also delays the signal. Keep DAW delay compensation enabled and change timing or quality modes while playback is stopped when possible.

Presets, history, and session recall

The header provides Previous, the current preset menu, Next, Save User Preset, Undo, and Redo. Factory presets are grouped as START, SOURCE, BUS, MASTER, and MOTION. The complete factory set is: Init, Gentle Balance, Broad Control, Vocal Focus, Bass Foundation, Drum Tamer, Mix Bus Balance, Drum Bus Glue, Mastering Touch, Kick Duck, Rhythmic Pump, Transient Lift, Transparent Master, Loudness Guard, De-Harsh Master, Low End Anchor, Stereo Focus, Guitar Smooth, Keys Space, and Parallel Drum Punch. User presets use the .tbmbc extension and are kept in Documents/TB_MultiBandCompressor/Presets/User. Presets and DAW sessions retain the audible band settings, global quality choices, and bypass state. A host's generic parameter view may also expose legacy compatibility controls such as global Output, per-band Output, detector coordinates, Detect, or Audition. The current custom panel intentionally hides those items: its detector follows the processing band, and DELTA is the intended band-change monitor.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
BAND ON / OFF	Turns the selected band slot on or off. An off band leaves its area on the direct signal path. Use the fewest active bands that solve the problem so the remaining spectrum stays untouched.
FREQUENCY	Places the center of the selected processing area. Sweep with DELTA briefly, then return to the full mix and place the band where the musical problem is clearest.
WIDTH	Sets how narrow or broad the selected frequency area is. Start broad enough to catch the event, then narrow it until adjacent tone stops moving unnecessarily.
THRESHOLD	Sets the event level where COMP or EXPAND begins to react. Set it while the representative passage loops and watch the red activity contour rather than chasing a fixed number.
RANGE	Limits the band's maximum change and chooses its direction: negative reduces and positive lifts. Keep the smallest movement that solves the problem and leave headroom when using positive lift.
RATIO	Sets how strongly the selected band moves after the detector enters the active region. Increase it only after Threshold and Range are working; Range remains the final ceiling on movement.
KNEE	Changes the transition around Threshold from firm toward gradual. Use a gentler onset for transparent control and a firmer onset when rhythmic definition is the goal.
ATTACK	Sets how quickly the selected band responds to a new event. Slow it enough to preserve useful punch, or speed it up when the front of the event is the problem.
RELEASE	Sets how quickly the selected band returns after the event settles. Set it to recover before the next event without obvious pumping or a band that never lets go.

Control	What it does and when to use it
COMP / EXPAND	Chooses COMP above Threshold or EXPAND below Threshold. Range still decides whether the result is reduction or lift. Decide which side of Threshold should react first, then use Range to decide up or down.
SIDECHAIN	Chooses the main input or the DAW auxiliary key as the selected band's detector source. Route and audition the DAW key before enabling it; all externally keyed bands share that one input.
STEREO / MID / SIDE	Chooses the complete Stereo image, centered Mid content, or Side width content for the band. Use Mid for centered weight or vocal focus and Side for ambience or width; always check mono after Side processing.
STEREO LINK	Coordinates left and right detector movement in Stereo mode. Keep it high for a stable image and lower it only when independent Stereo movement is musically useful.
DYNAMIC / LINEAR	Chooses immediate DYNAMIC processing or the more latent LINEAR phase path. Begin with DYNAMIC and move to LINEAR only when its phase behavior clearly improves the source.
OVERSAMPLING	Chooses additional internal quality for fast gain movement at the cost of CPU and delay. Raise it after the sound is set and keep the lightest option that removes an audible problem.
LOOKAHEAD	Delays the signal so the detector can prepare for sudden peaks. Use only enough to catch the peak and change it while playback is stopped when the host handles latency changes poorly.
MIX	Blends the aligned direct signal with all active band processing. Use it for parallel control after the individual bands are behaving correctly.
BYPASS	Returns the latency-aligned direct signal for comparison. Compare the same compensated passage at a similar loudness and let parameter smoothing settle.

Practical uses

Focused vocal control

- Start with Vocal Focus and loop a phrase containing body, presence, and a difficult consonant.
- Use one broad Mid band for vocal body and a narrower Mid band for the harsh or sibilant area.
- Set negative Range for the problem band and use a fast enough Attack to catch it without flattening every consonant.
- Use DELTA to confirm that the band is not removing the vowel or room tone, then judge the vocal in the mix.

Expected result: The vocal stays centered and readable while only the distracting frequency events move back.

Bass foundation or kick ducking

- Use Bass Foundation for internal control, or Kick Duck when a DAW sidechain key is routed.
- Place a Mid band over the competing low-frequency area and enable SIDECHAIN only when an external key is connected.
- Set negative Range and adjust Threshold until the red contour moves only on the intended key events.
- Tune Attack and Release to the groove, then reduce Mix if the movement should feel more parallel.

Expected result: Low end becomes more stable or makes space rhythmically without pulling the whole mix down.

Drum punch and harshness

- Start with Drum Tamer, Drum Bus Glue, Transient Lift, or Parallel Drum Punch.
- Use separate bands for weight, boxiness, attack, and cymbal edge instead of making every band equally strong.
- Use positive Range with suitable timing to lift attack, or negative Range to contain an aggressive area.
- Check DELTA, then return to the full signal and use Mix to preserve natural punch.

Expected result: The drum bus gains controlled impact while low, mid, and high events keep independent movement.

Mix or master balance

- Begin with Mix Bus Balance, Mastering Touch, or Transparent Master and keep the number of active bands small.
- Use broad, gentle bands and modest Range so the spectrum breathes rather than sounding re-equalized.
- Use Mid for centered weight and Side only when ambience or width needs separate control.
- Compare DYNAMIC and LINEAR at equal loudness and keep the lighter mode unless the more expensive path provides a clear benefit.

Expected result: The bus feels steadier and more finished without obvious full-band pumping or unnecessary spectral splitting.

Stereo motion and sound design

- Load Rhythmic Pump, Keys Space, or Stereo Focus as a directional starting point.
- Use positive or negative Range with EXPAND around quieter events, targeting Side for animated width or Mid for a moving center layer.
- Check mono, leave output headroom, and save the result as a User preset after DELTA is off.

Expected result: The source develops frequency-selective motion and space while the untouched spectrum remains connected.

Common pitfalls

If a band seems to do nothing, confirm that it is enabled, Threshold reaches the relevant events, Range is not neutral, and DELTA is off after checking.

COMP or EXPAND does not by itself decide whether a band becomes louder or quieter; the sign of Range decides the direction.

If SIDECHAIN is enabled without a routed auxiliary input, the panel reports NO SC INPUT and the detector falls back to the main signal.

A mono external key contains no Side information, so a Side-targeted band may receive no useful detector movement from that key.

Overlapping broad bands and strong positive Range can consume headroom even when each band sounds reasonable in isolation.

LINEAR processing can introduce pre-ringing, while high Oversampling and Lookahead increase CPU use and latency.

The analyzer shows input only. Use the red contour, DELTA, meters, and your ears to judge processing rather than looking for an output spectrum.

The plug-in has four band slots and one shared external sidechain bus; it does not provide a different DAW key input for every band.